

Non-Preemptive Shortest Job First CPU Scheduling Algorithm

Completed

Write a C program to implement the Non-Preemptive Shortest Job First (SJF) CPU Scheduling Algorithm. The program should accept the number of processes, their Arrival Times (AT), and Burst Times (BT). At any given time, the process with the minimum burst time among the arrived processes must be executed first. Calculate the Completion Time (CT), Turnaround Time (TAT), and Waiting Time (WT) for each process and find the overall average TAT and WT.

Core Logic

Selection: Out of all processes that have arrived by the current time, the one with the minimum Burst Time is chosen.

Formulae:

- $\text{Turnaround Time (TAT)} = \text{Completion Time (CT)} - \text{Arrival Time (AT)}$
- $\text{Waiting Time (WT)} = \text{Turnaround Time (TAT)} - \text{Burst Time (BT)}$

Select Language : C (GCC 9.2.0)

```

1 #include<stdio.h>
2 int main(){
3     int n;
4     printf("Enter number of processes: \n");
5     scanf("%d",&n);
6     int a[n];
7     int b[n];
8     int c[n];
9     int t[n];
10    int w[n];
11    int time=0;
12    int done[n];
13    int completed=0;
14    int min;
15    int idx;
16    int g=1;
17    for(int i=0;i<n;i++){
18        printf("Enter Arrival Time and Burst Time for P%d",i);
19        scanf("%d %d",&a[i],&b[i]);
20        g++;
21        done[i]=0;
22    }
23    printf("\n");
24    while(completed<n){

```

Test Cases



Run Sample Cases



Run All Cases

Custom Test Case

> Custom Test

Sample Test Case

> Test Case - 1

> Test Case - 2